

The Lifetime Incidence of Transfer Programs

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→ Ambiguous whether transfers are more progressive in lifetime or current terms

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- More reasonable to define social welfare function over lifetime earnings
 - Without frictions, lifetime earnings = consumption → marginal utility of income (Modigliani & Brumberg 1954, Friedman 1957, MaCurdy 1981)
 - Current income is an unsatisfactory measure of consumption or material hardship (Meyer & Sullivan 2003, Sullivan et al. 2008)

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- In total: \$4500 vs. \$2200 per person-year at bottom of lifetime vs. current distribution
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③ Evaluate efficiency gains from transfers versus nonlinear income tax

- Benefits depend on lifetime earnings, not just current $\rightarrow$ gains from tagging
- Find about 50% smaller efficiency cost of transfers than “equivalent” taxes

Literature

- **Lifetime incidence of taxes and government policies**

Davies et al. 1984, Bane & Ellwood 1986, Poterba 1989/1991, Slemrod 1992, Fullerton & Lim 1993, Lyon & Schwab 1995, Bengtsson et al. 2016, Catherine & Yannelis 2020

→ first analysis of lifetime incidence with realistic heterogeneity or incomplete take-up

- **Targeting properties of transfer programs**

Parsons 1991, Dynarski & Scott-Clayton 2006, Bhargava & Manoli 2015, Alatas et al. 2016, Deshpande & Li 2019, Finkelstein & Notowidigdo 2019, Gray 2019, Stepner 2019

→ new decomposition offers a precise & generalizable meaning to claims that eligibility rules or selective take-up improve or harm targeting

- **Efficient taxation & transfer programs**

Akerlof 1978, Nichols & Zeckhauser 1982, Stern 1982, Stiglitz 1982, Diamond & Sheshinski 1995, Coate 2000, Saez 2002, Werning 2007, Kleven & Kopczuk 2011, Hendren 2020

→ money-metric measure of efficiency gain from “tagging”

Outline

- ① Data and Measurement
- ② Current and Lifetime Incidence
- ③ Decomposition
- ④ Implications for Efficient Redistribution

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Panel Survey of Income Dynamics (PSID) waves, 1997–2019

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Other Data Sources

- Simulated eligibility using PSID + program rules
- CPS: Impute cash-equivalent values of Medicaid, WIC, and housing assistance

Estimating Lifetime Earnings

We estimate a panel data model of individual earnings by Poisson PML:

$$\log E[y_{it} | x_{it}] = \alpha_i + \alpha_t + x'_{it}\beta$$

where

- α_i : individual FE
- α_t : year FE
- x_{it} : time-varying covariates (age, marital status, children, education)

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Steps to measure lifetime earnings

[Details](#)[Validation](#)

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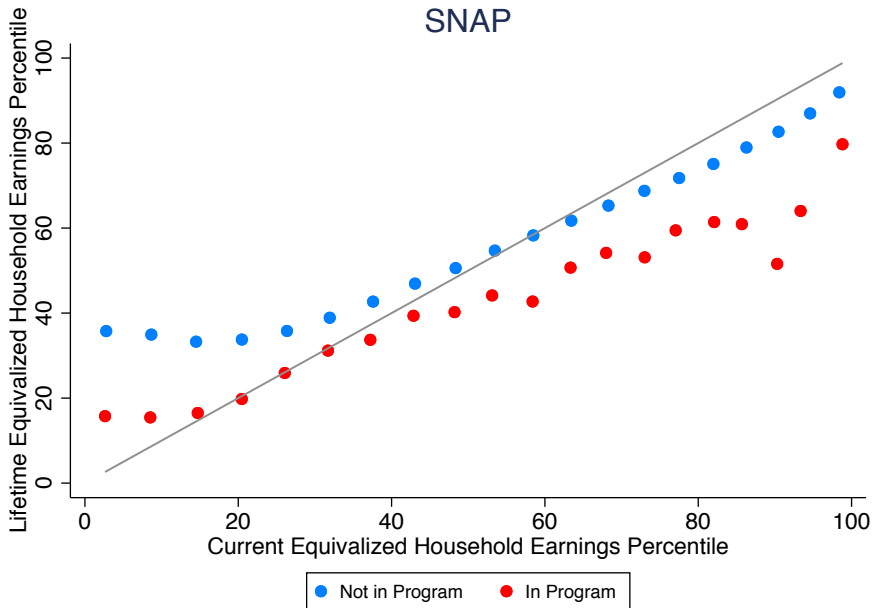
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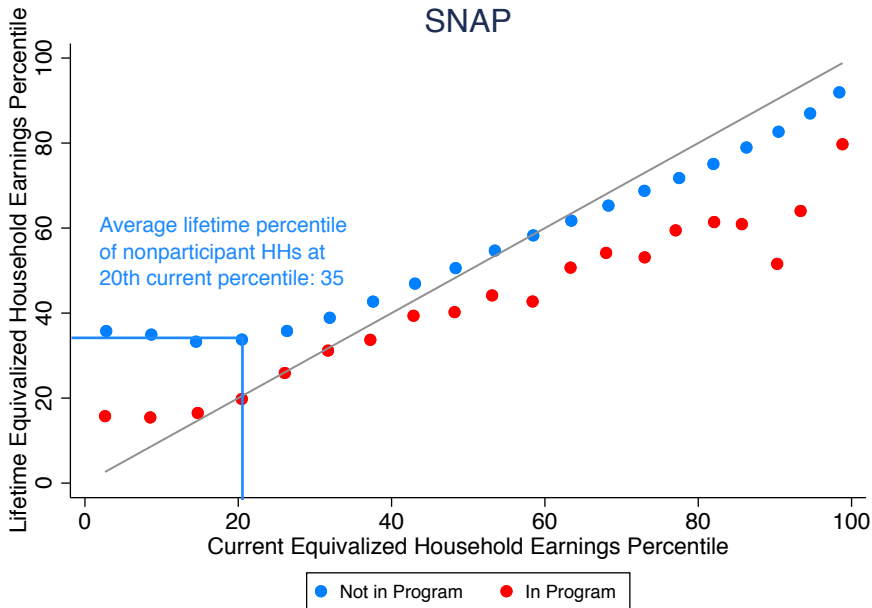
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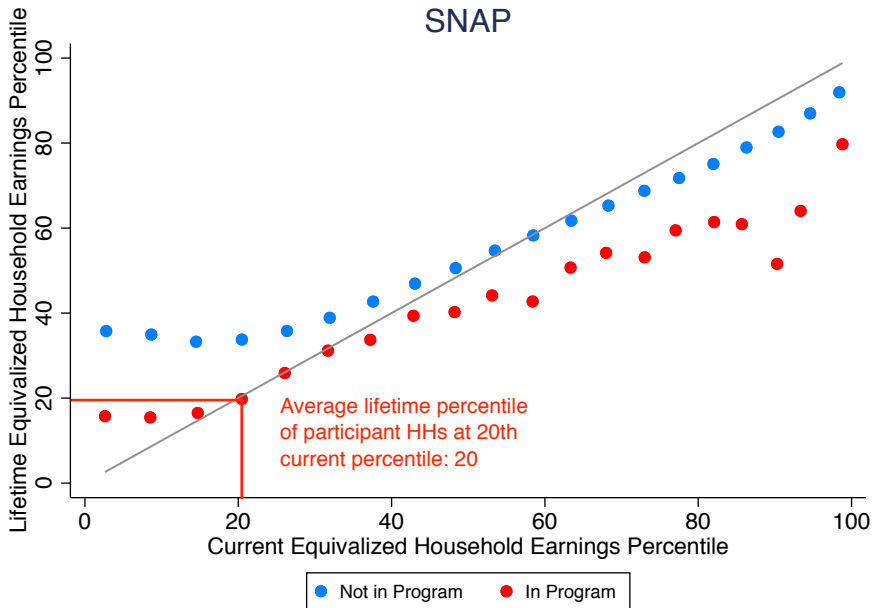
- 1 Shrink estimated FEs $\{\hat{\alpha}_i\}$ by Empirical Bayes
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- 3 For households: Add up predicted earnings and apply equivalence scale



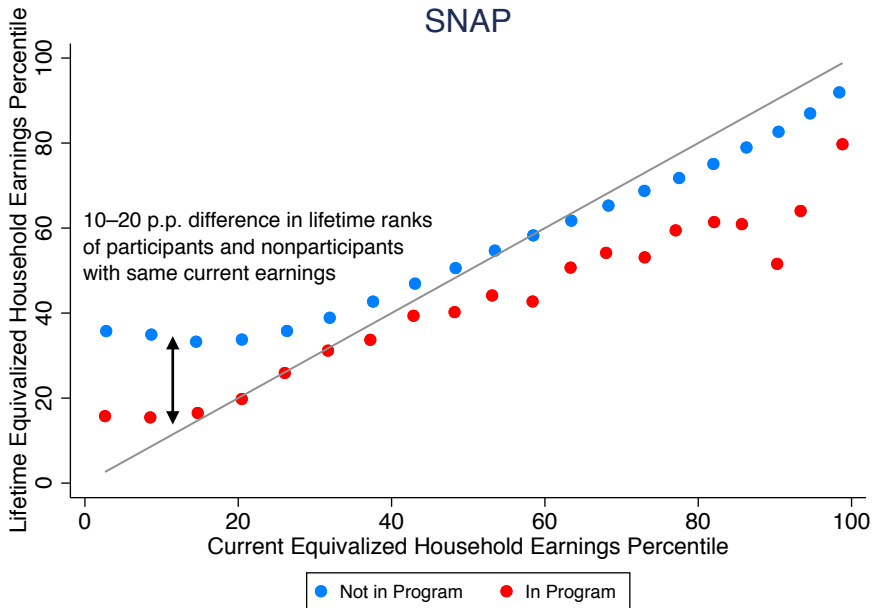
SNAP

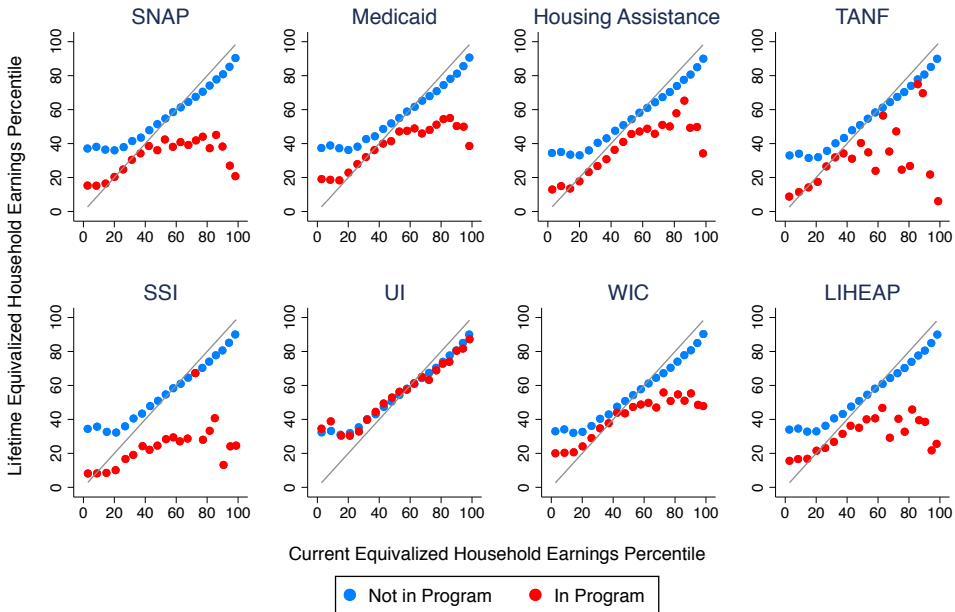


SNAP



SNAP




[More](#)

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- ① Data and Measurement
- ② **Current and Lifetime Incidence**
- ③ Decomposition
- ④ Implications for Efficient Redistribution

Who Benefits from Means-Tested Transfers?

We study incidence graphically by local linear regressions of outcomes on earnings ranks.

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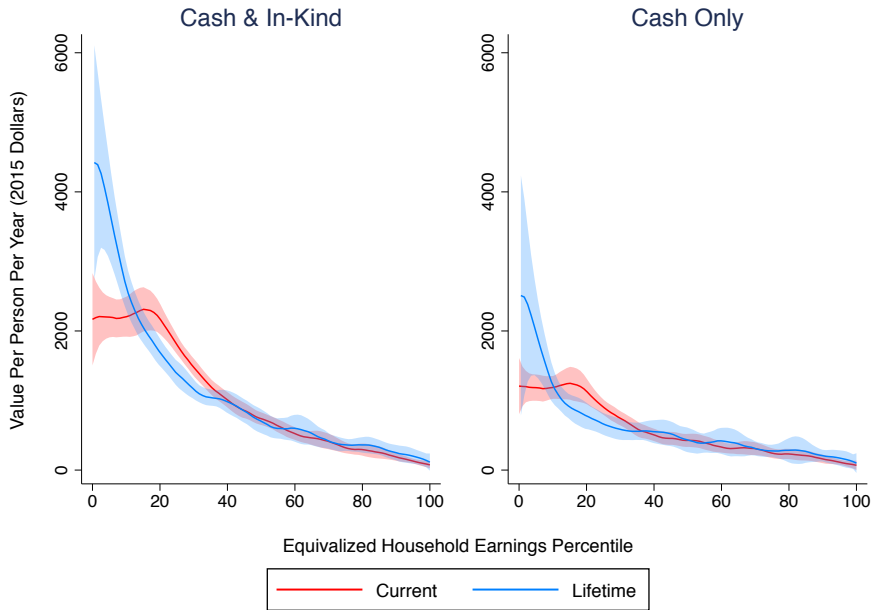
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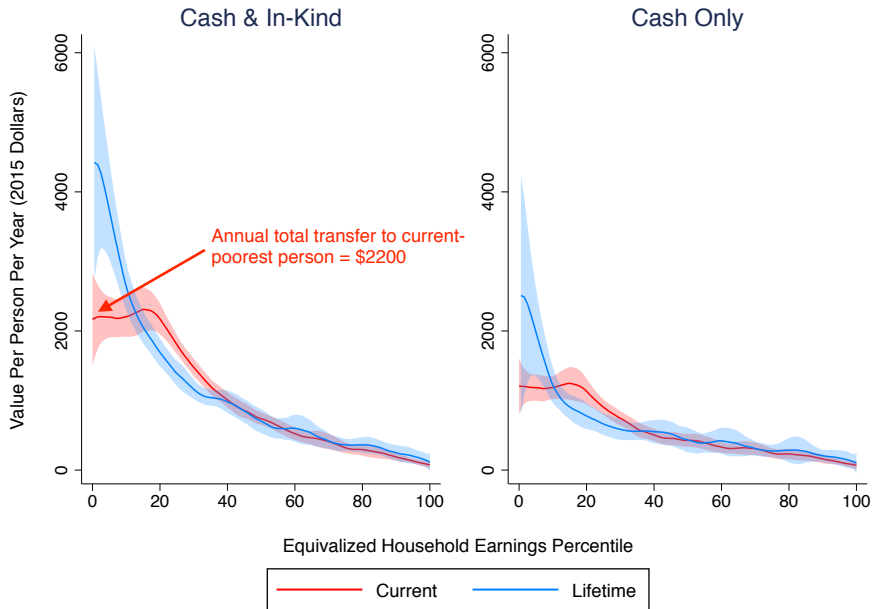
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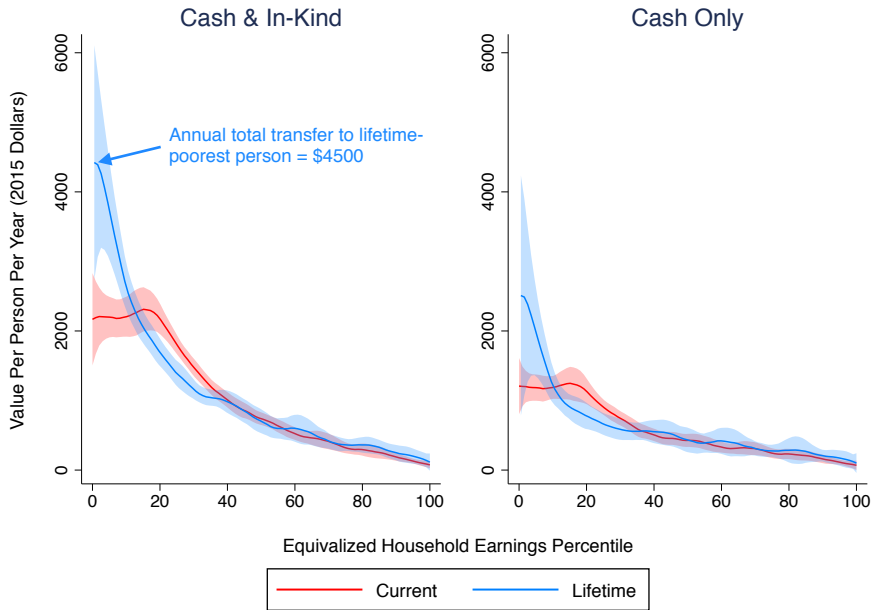
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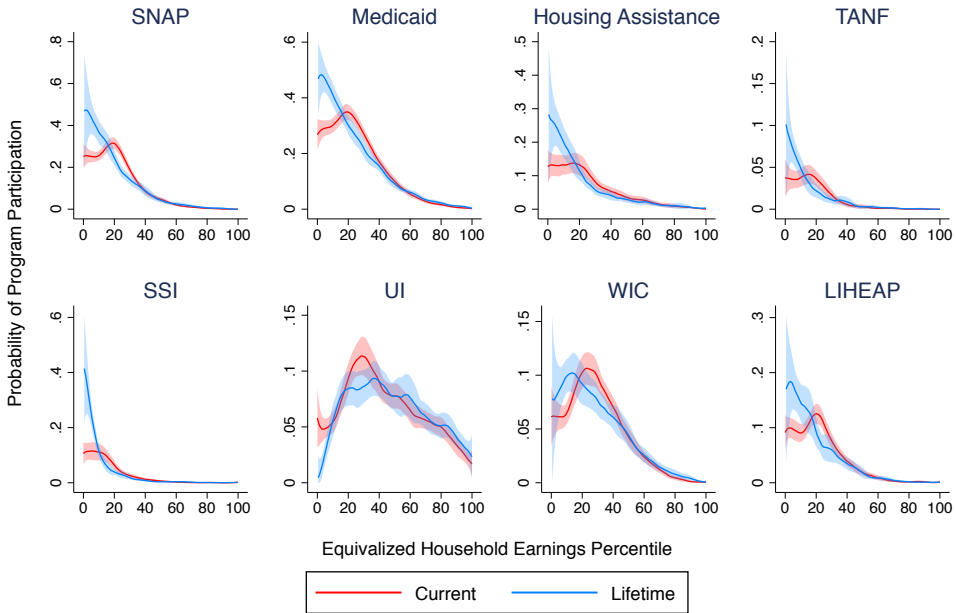
Throughout looking at current and lifetime earnings ranks of:

- Individuals
- Equivalized households
- Unequivalized households





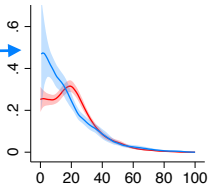




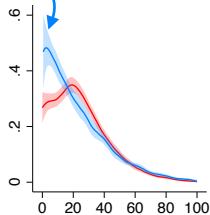
More

Participation rate in
SNAP & Medicaid by
lifetime poorest: 50%

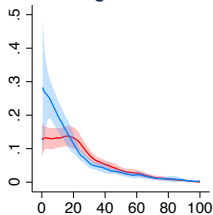
Probability of Program Participation



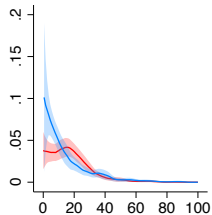
Medicaid



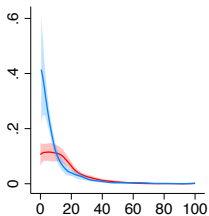
Housing Assistance



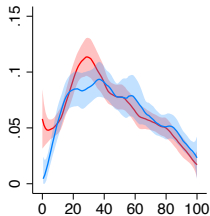
TANF



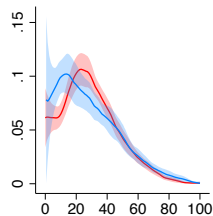
SSI



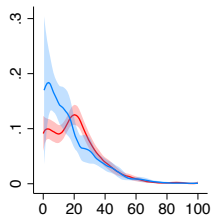
UI



WIC



LIHEAP

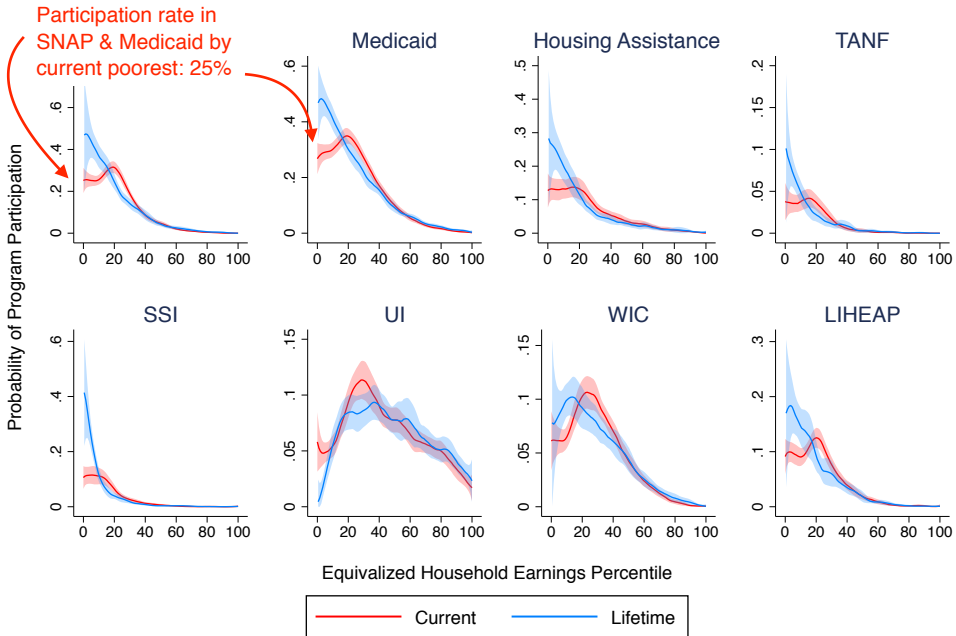


Equivalized Household Earnings Percentile

— Current

— Lifetime

More



More

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Mobility, Eligibility, and Take-Up Effects

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- 2 **Eligibility:** Screening criteria other than current earnings may have residual predictive power for lifetime earnings
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→ Can be applied to program participation and per-capita annual value of transfers

Decomposition

Counterfactual 1 (mobility effect only): Lifetime incidence if participation were purely a function of current earnings rank, then run through lifetime mobility matrix

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$$\widetilde{p}_M(r) = \int \Pr(D_{it} = 1 \mid r_{it}) dF(r_{it} \mid \bar{r}_i = r)$$

where

- $\Pr(D_{it} = 1 \mid r_{it})$: participation rate at current rank r_{it}
- $F(r_{it} \mid \bar{r}_i = r)$: conditional distribution of current ranks r (given lifetime rank $\bar{r}_i$)

Decomposition

Counterfactual 2 (mobility + eligibility effects): Lifetime rank matters only via eligibility, thus no selective take-up among eligibles on lifetime rank

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$$\widetilde{p}_E(r) = \int \Pr(D_{it} = 1 \mid r_{it}, E_{it} = 1) \Pr(E_{it} = 1 \mid r_{it}, \bar{r}_i = r) dF(r_{it} \mid \bar{r}_i = r)$$

where

- $\Pr(D_{it} = 1 \mid E_{it} = 1, r_{it})$: take-up rate among eligibles at current rank r_{it}
- $\Pr(E_{it} = 1 \mid r_{it}, \bar{r}_i = r)$: eligible share given current rank r_{it} and lifetime rank $\bar{r}_i$

Decomposition

Let $p_L(r) = \Pr(D_{it} = 1 \mid \bar{r}_i = r)$ and $p_C(r) = \Pr(D_{it} = 1 \mid r_{it} = r)$. The decomposition is:

$$p_L(r) - p_C(r) = \underbrace{[p_L(r) - \widetilde{p}_E(r)]}_{\text{take-up effect}} + \underbrace{[\widetilde{p}_E(r) - \widetilde{p}_M(r)]}_{\text{eligibility effect}} + \underbrace{[\widetilde{p}_M(r) - p_C(r)]}_{\text{mobility effect}}.$$

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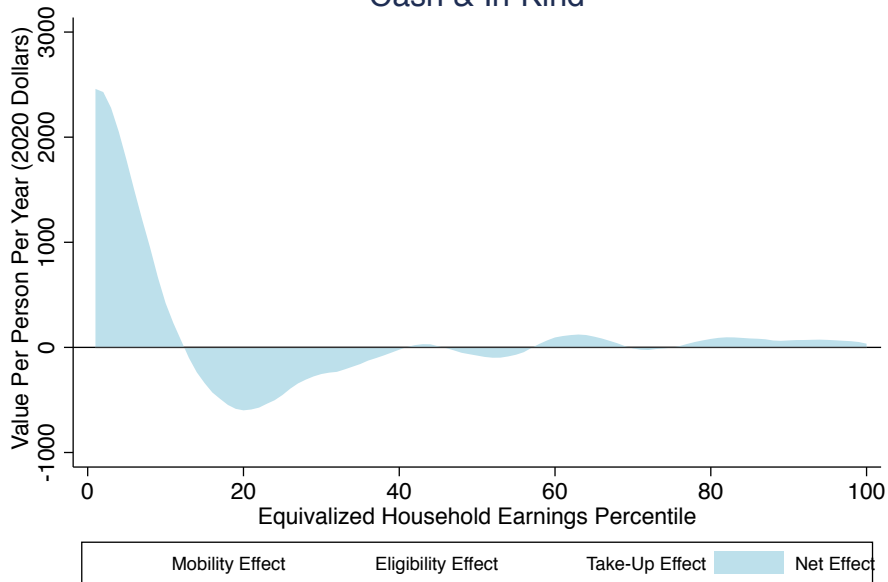
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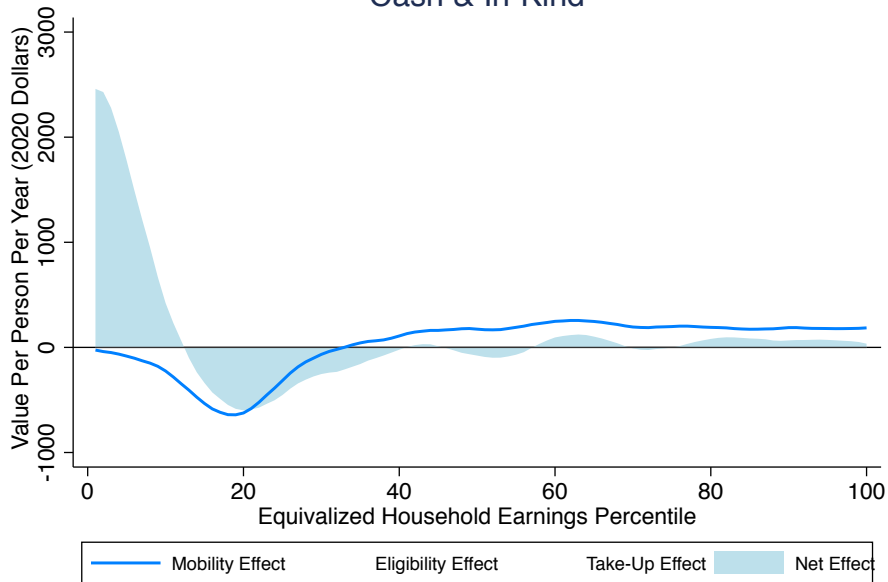
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- **Take-up effect:** Actual lifetime incidence vs. lifetime incidence in counterfactual in which participation is informative only via eligibility

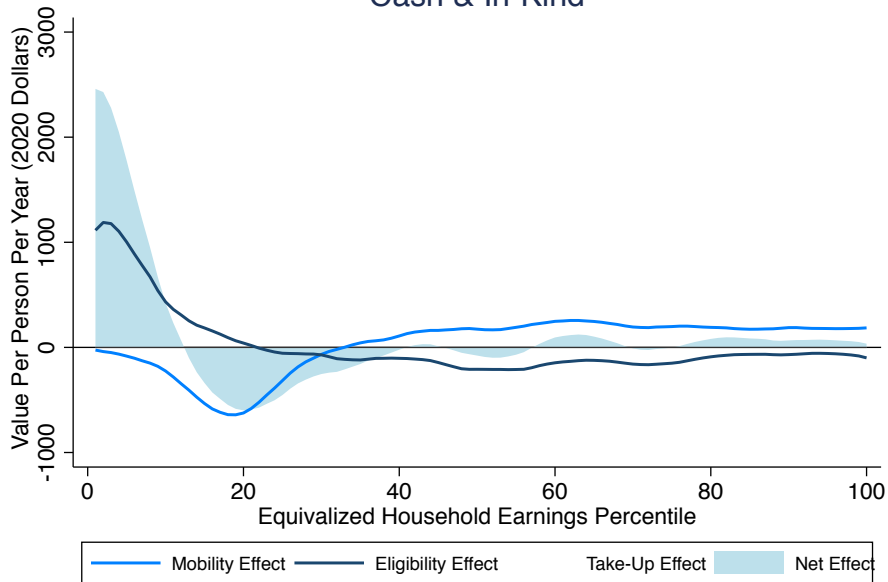
Cash & In-Kind



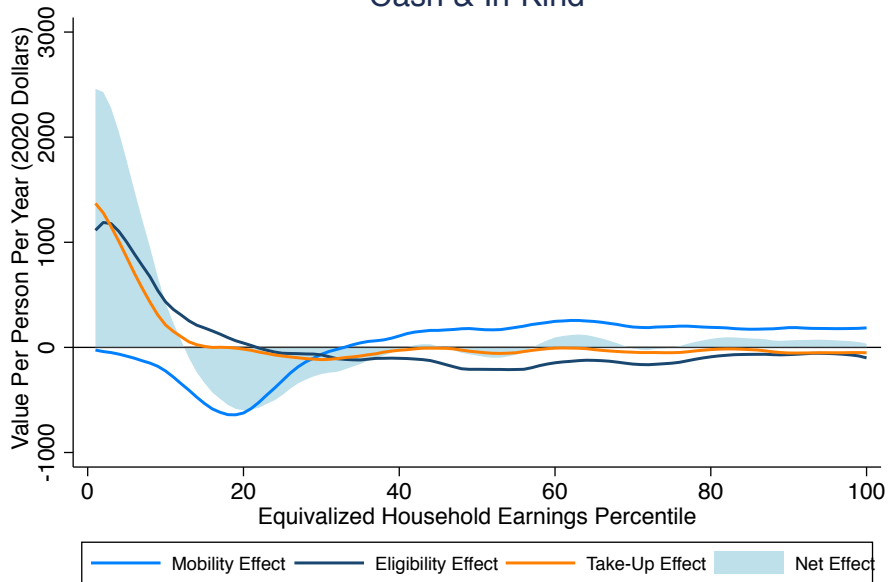
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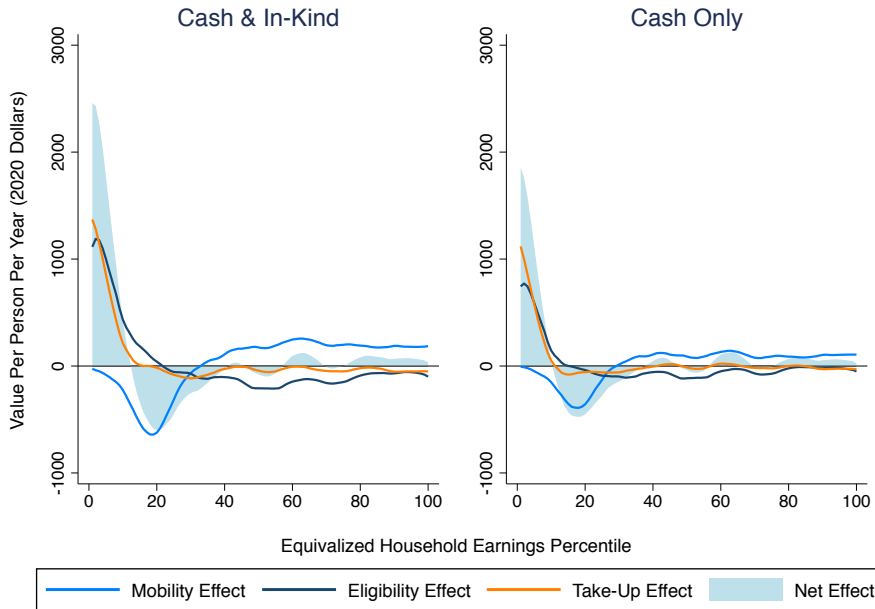


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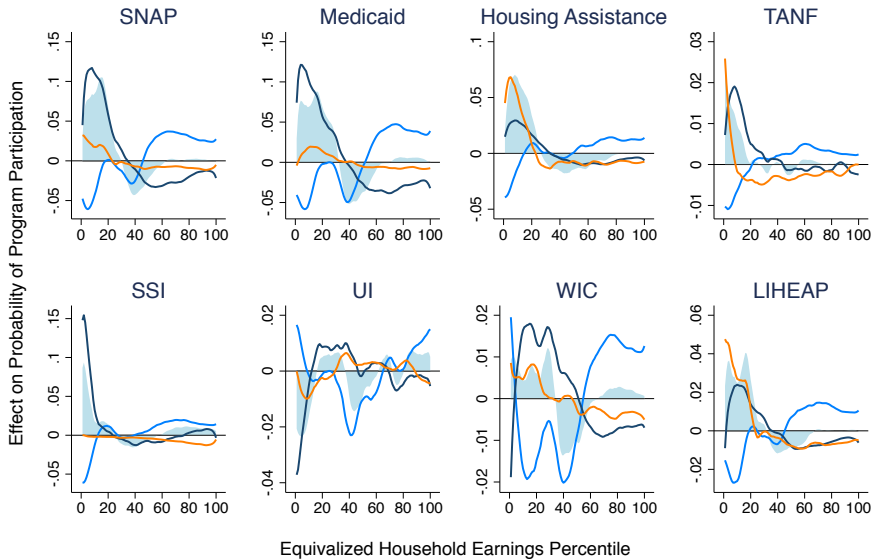


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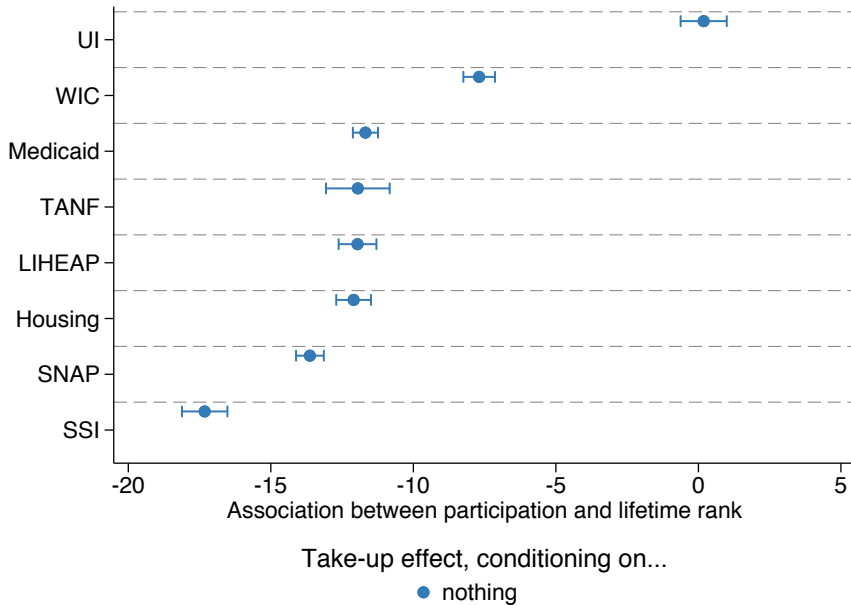


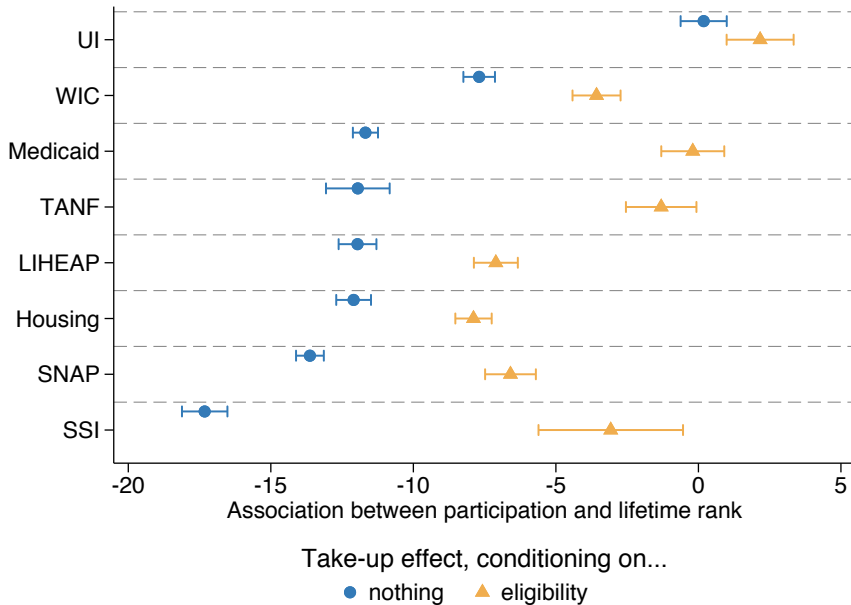
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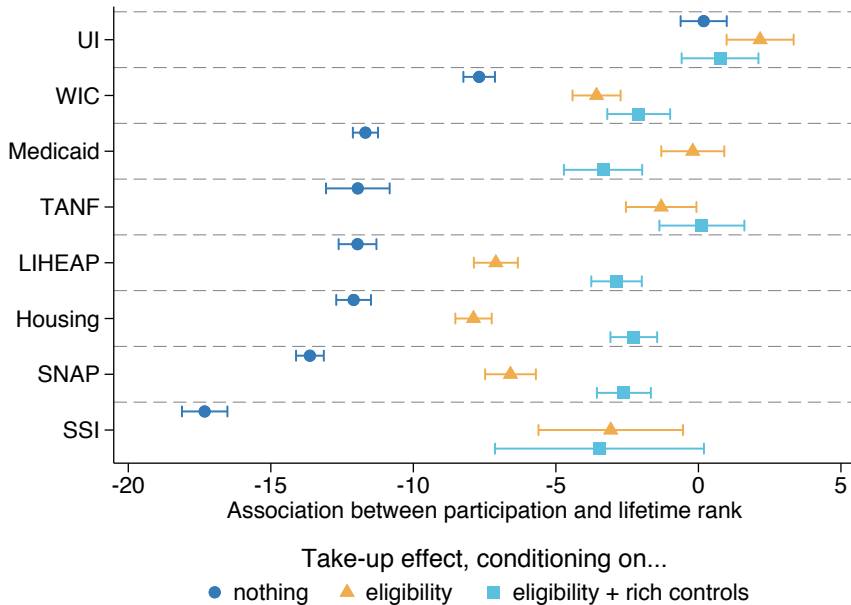


— Mobility Effect
 — Eligibility Effect
 — Take-Up Effect
 Net Effect

[More](#)







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- ③ Decomposition
- ④ **Implications for Efficient Redistribution**

Measuring Efficiency Gains from Transfers Versus Taxes

Consider a small reform involving a transfer program $t(y, \bar{y})$ and tax schedule $T(y)$

- 1 Increase transfer program proportionally by \$1

$$\Delta t(y, \bar{y}) = \epsilon t(y, \bar{y})$$

- 2 Implement change in taxes that neutralizes impact on lifetime income distribution

$$\Delta T(y) \quad \text{such that} \quad \int [\Delta T(y) - t(y, \bar{y})] dH(y|\bar{y}) = 0 \quad \text{for all } \bar{y}$$

Efficiency metric: What is the net fiscal externality of this reform?

→ Measure local fiscal externalities following Hendren (2020) [More](#)

Fiscal Externalities from Transfers Versus Taxes

Total fiscal externality of transfer change:

$$FE_{\text{transfer}} = \int FE(y) \Delta t(y, \bar{y}) dH(y, \bar{y})$$

Total fiscal externality of tax change:

$$FE_{\text{tax}} = \int FE(y) \Delta T(y) dH(y)$$

Net fiscal externality:

$$\Delta FE = FE_{\text{tax}} - FE_{\text{transfer}}$$

If $\Delta FE > 0$, this reform saves government money but leaves utilities unchanged

→ Monetizes the gains from capacity of transfers to condition on $\bar{y}$

Fiscal Externalities from Transfers Versus Taxes

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- Generalizes and quantifies efficiency argument in Diamond & Sheshinski (1995)

Possible generalizations

- Incorporate elasticity of lifetime income (→ not truly lump sum)
- Incorporate heterogeneous fiscal externalities (e.g., b/c of transfer programs)

Efficiency Gains from Transfer Programs

Program	FE_{tax}	$FE_{transfer}$	ΔFE	$\Delta FE / FE_{tax}$
All Transfers	0.051	0.028	0.023	0.46
All Cash Transfers	0.033	0.021	0.011	0.34
SNAP	0.083	0.044	0.039	0.47
Medicaid	0.075	0.037	0.039	0.52
TANF	0.153	0.032	0.121	0.79
SSI	0.051	0.030	0.021	0.42
Housing Assistance	0.063	0.035	0.028	0.45
WIC	0.112	0.110	0.002	0.02
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 - Efficiency gains from tagging are large in percentage terms
 - Small in dollar terms because calibrated fiscal externalities are small at bottom

Backup Slides

Step 1: Empirical Bayes Shrinkage of Estimated Fixed Effects

Following Morris (1983), we shrink estimated FEs towards conditional mean predicted by time-invariant covariates (Jacobs & Lefgren 2007, Chandra et al. 2016, Sorkin 2017)

$$\alpha_i^* = \frac{\sigma_i^2}{\sigma_i^2 + \tau^2} \cdot z_i' \lambda + \frac{\tau^2}{\sigma_i^2 + \tau^2} \cdot \hat{\alpha}_i$$

where

- z_i : time-invariant covariates (sex, race)
- σ_i^2 : sampling variance of $\hat{\alpha}_i$ (sample analog obtained via bootstrap)
- τ^2 : variance of true FEs α_i

→ Procedure iteratively updates $\hat{\lambda}, \hat{\tau}$ until convergence

Back

Steps 2 and 3: Fill in Panel & Equivalize Household Earnings

Step 2. Impute covariates & predicted earnings if not observed at ages from 18 to 65

→ Assume that x_{it} unchanged at latest observed value

$$\widehat{y}_{it}^* = \exp \left(\widehat{\alpha}_i^* + \widehat{\alpha}_{ct} + x_{it} \widehat{\beta} \right)$$

Step 3. For households, add up earnings and equivalize:

$$\widehat{y}_{ht}^* = f \left(\widehat{y}_{i(h)t}^* + \widehat{y}_{j(h)t}^* \right)$$

where

- $i(h)$ and $j(h)$: head and spouse of household h
- $f(\cdot)$: equivalence scale function (Meyer & Sullivan 2008)

Equivalence Scale Details

$$y_{ht} = \frac{1}{\psi_h} \sum_{i \in h} y_{it}$$

Options:

- Two-parameter family

$$\psi_h = (N_h^{\text{adults}} + \beta \cdot N_h^{\text{children}})^{\gamma}$$

$$\beta = 0.7, \gamma = 0.7$$

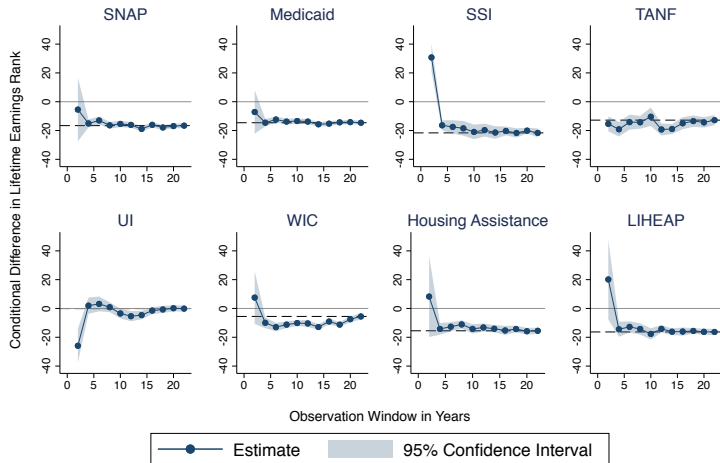
- Square-root

$$\psi_h = \left(N_h^{\text{adults}} + N_h^{\text{children}} \right)^{1/2}$$

- OECD

$$\psi_h = \left(1 + 0.5 \cdot (N_h^{\text{adults}} - 1) + 0.3 \cdot N_h^{\text{children}} \right)$$

Stable Predictive Effect of Transfers on Lifetime Rank



$$\bar{r}_i = \alpha + \beta r_{it} + \gamma D_{it} + e_i$$

Earnings Percentile Rank Definitions

Individual current earnings rank

$$r_{it} = \Pr(y \leq y_{it})$$

Household current earnings rank

$$r_{ht} = \Pr(y \leq y_{ht})$$

Individual lifetime earnings rank

$$\bar{r}_i = \Pr\left(y \leq T^{-1} \sum_t \hat{y}_{it}^*\right)$$

Household lifetime earnings rank

$$\bar{r}_h = \Pr\left(y \leq T^{-1} \sum_t \hat{y}_{ht}^*\right)$$

Misreporting in the PSID

- 10–30% of transfer recipients in PSID do not report assistance (Meyer et al. 2009)
 - Adjust for average reporting rates by program → makes our results larger
 - Relevant margin is participation, not amount conditional on participation

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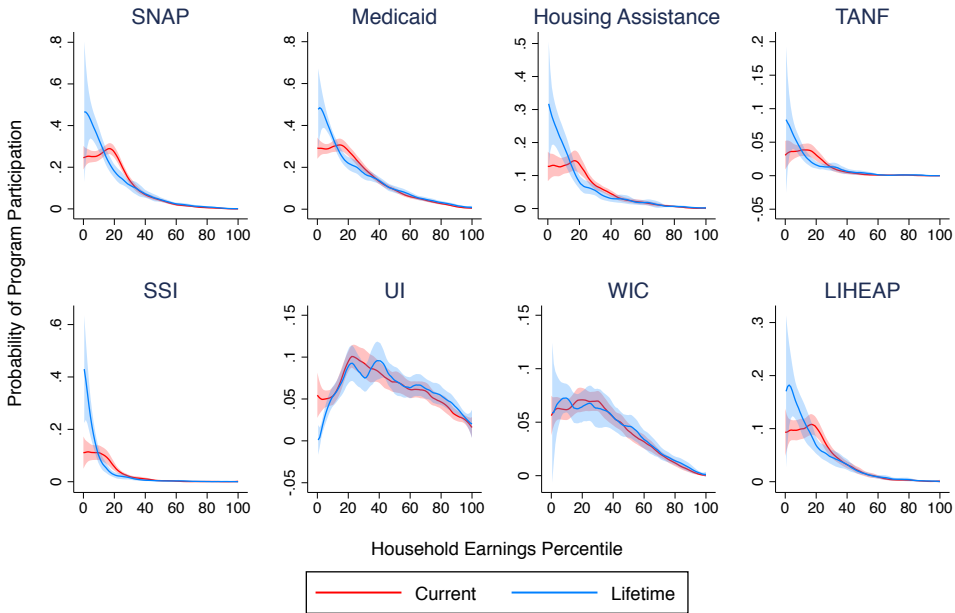
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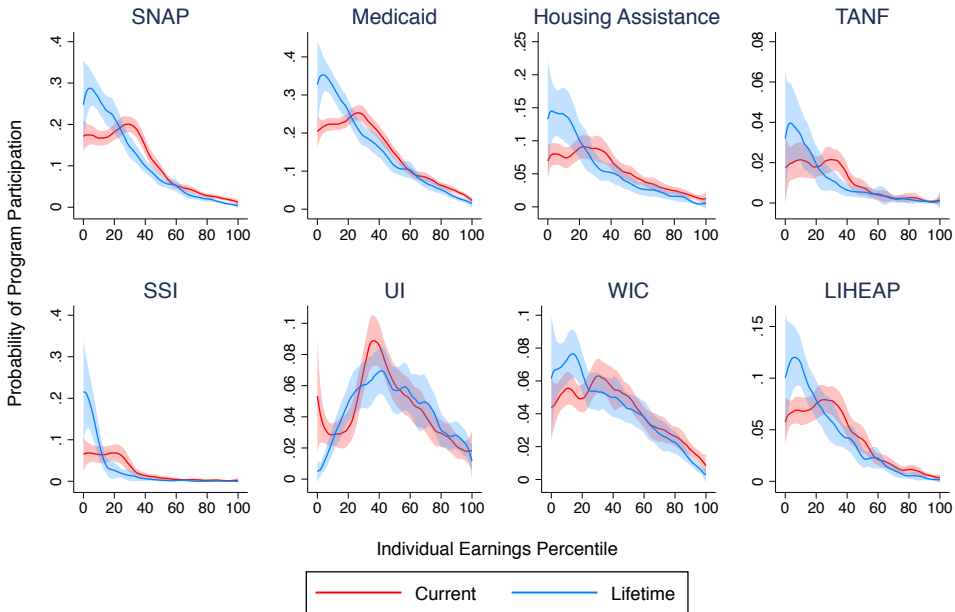
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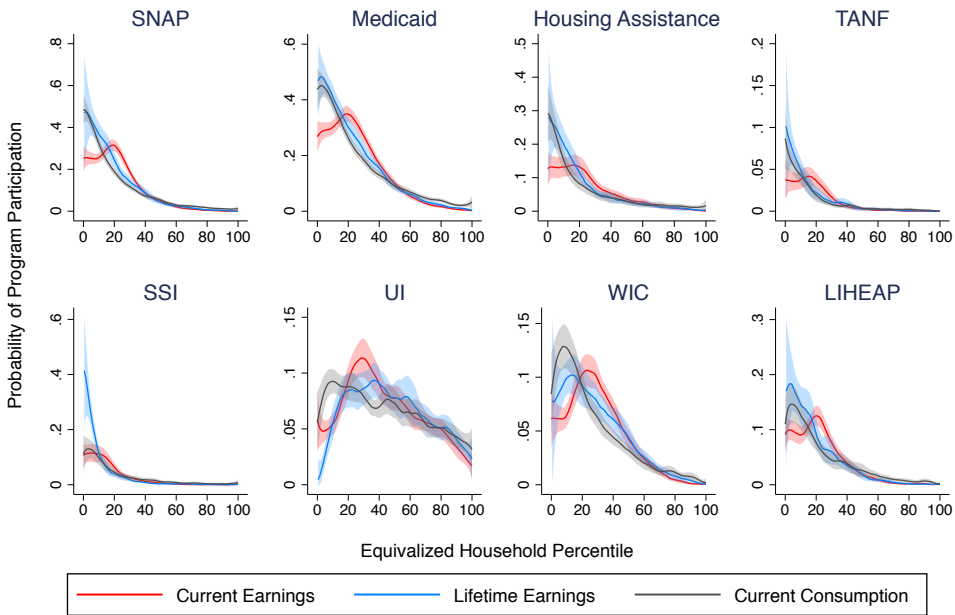
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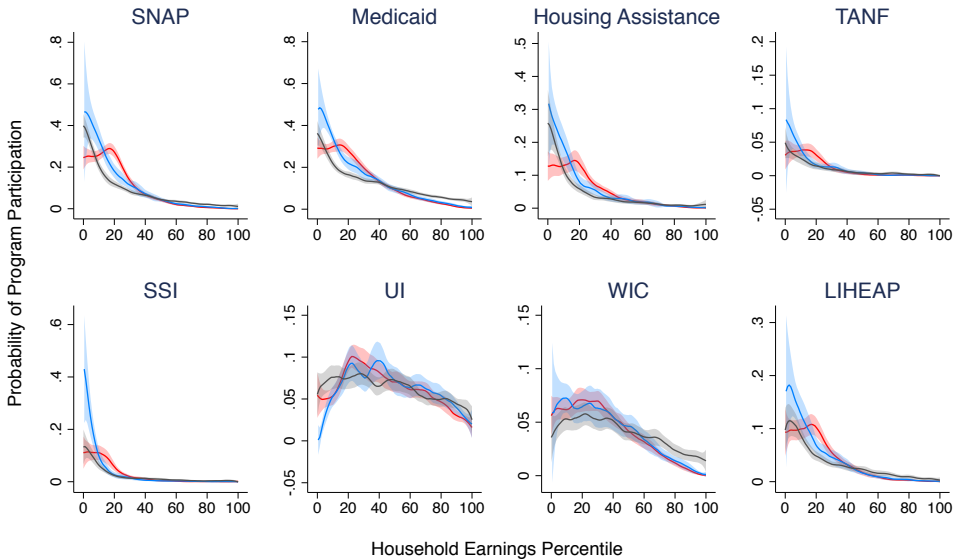
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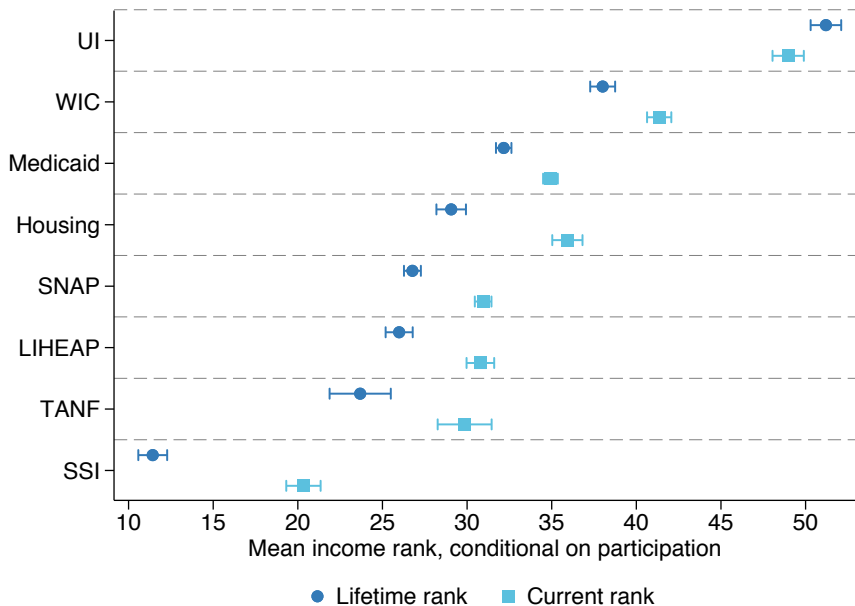
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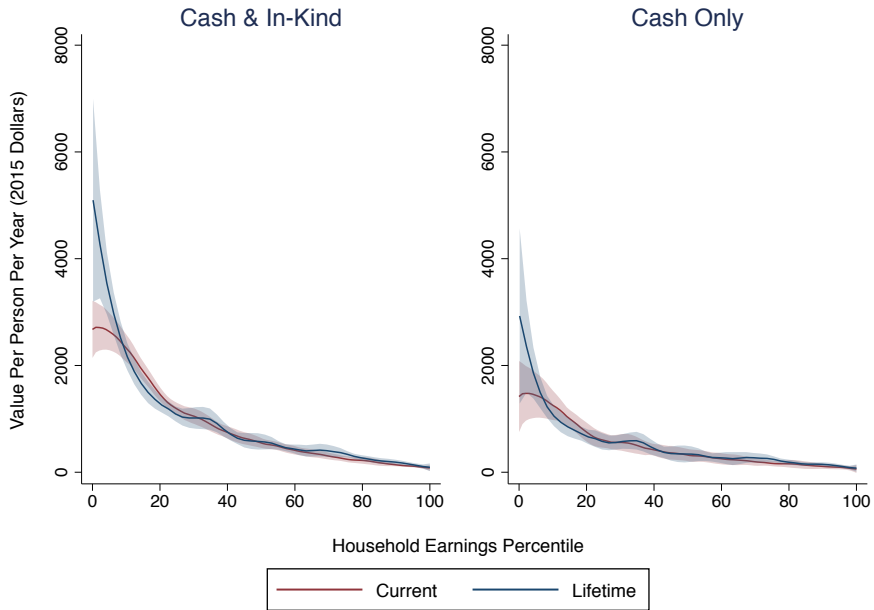


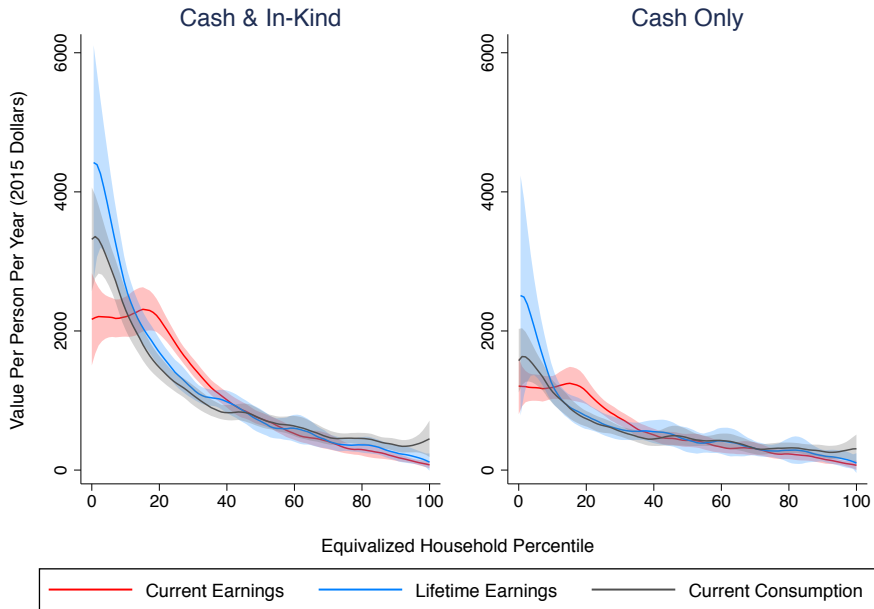


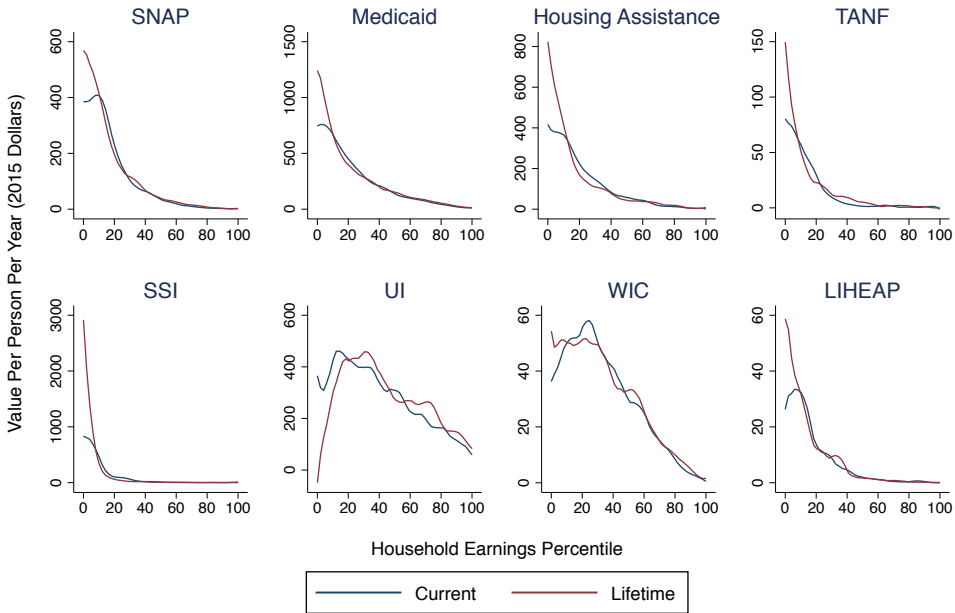


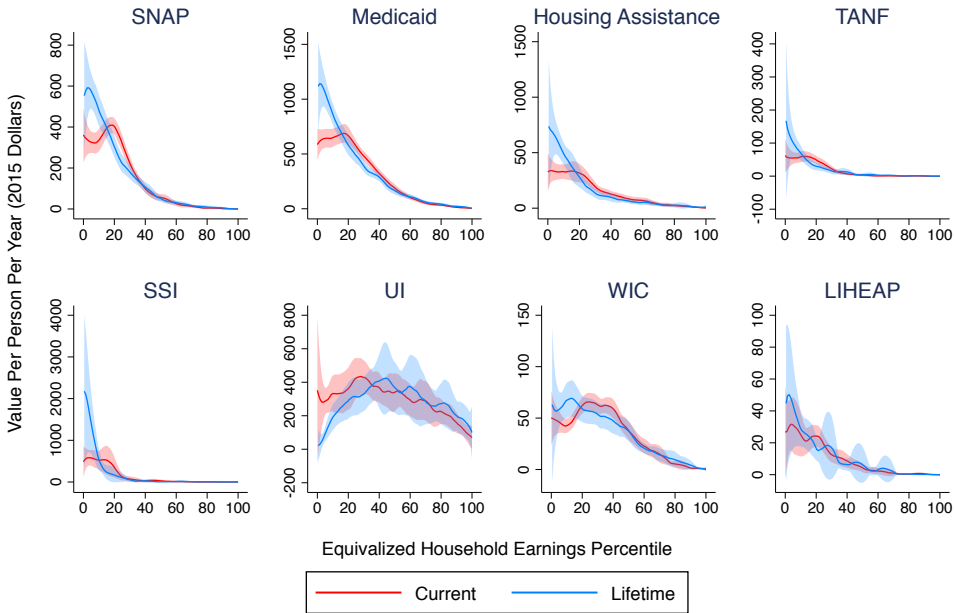


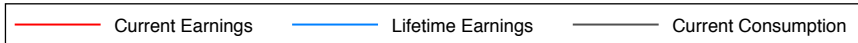
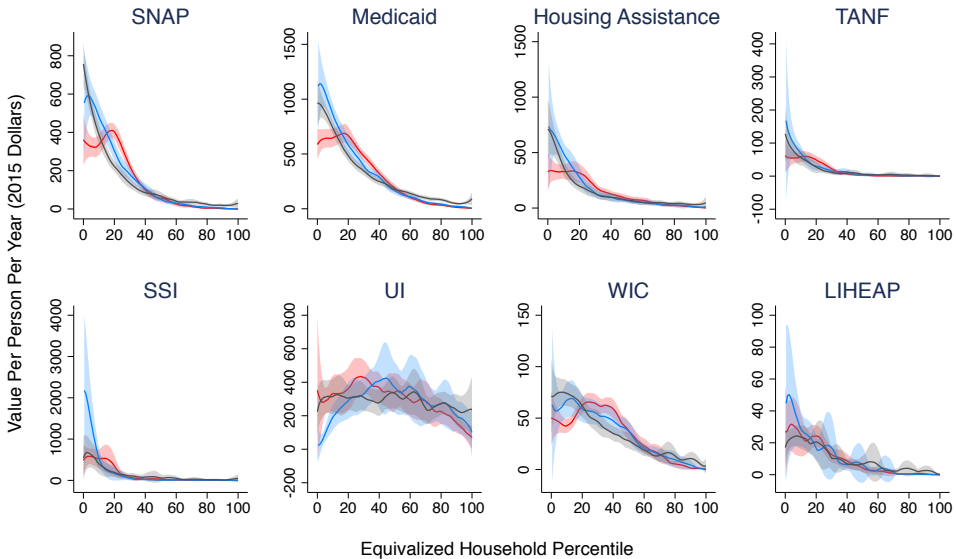


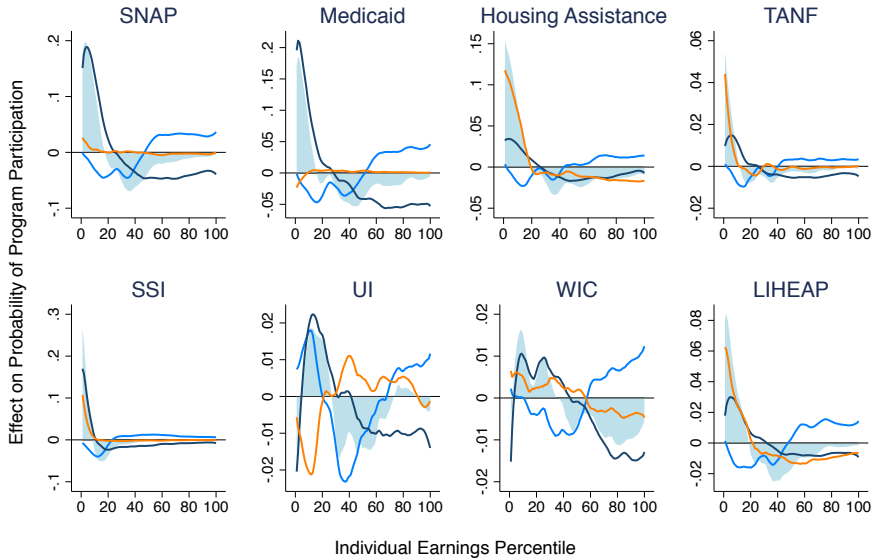






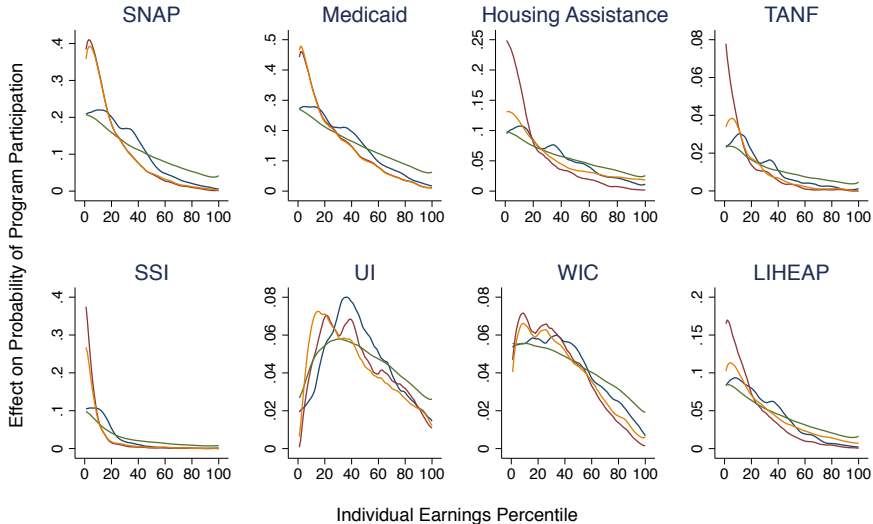


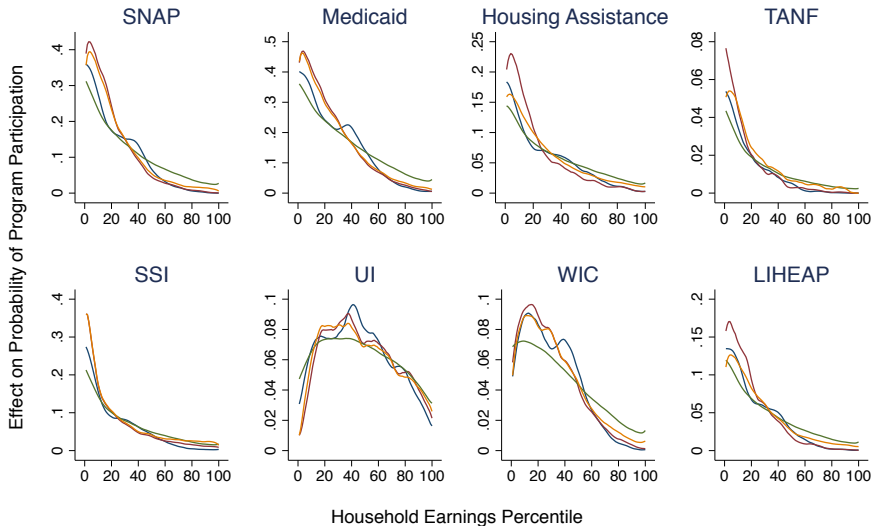


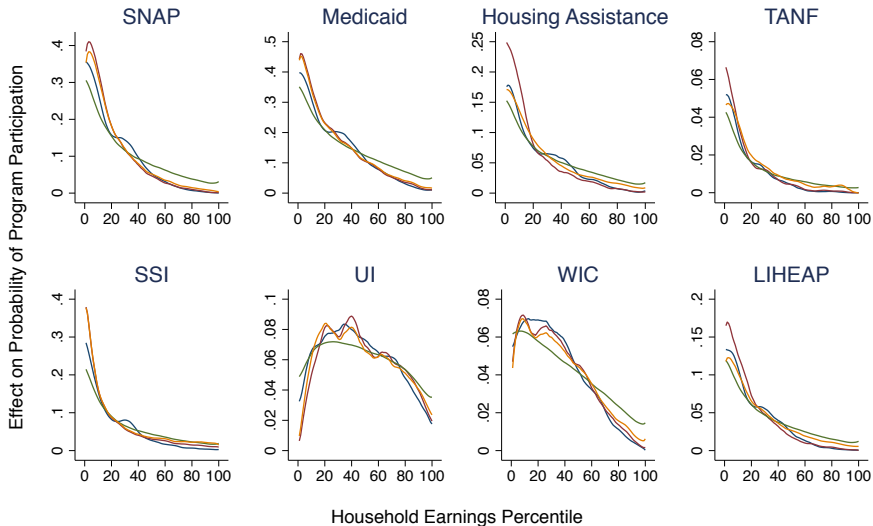


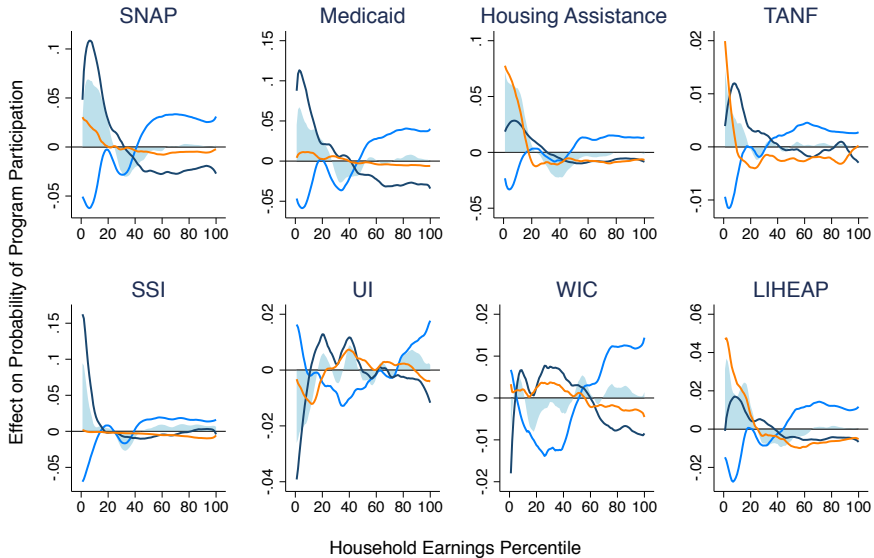
— Mobility Effect
 — Eligibility Effect
 — Take-Up Effect
 Net Effect

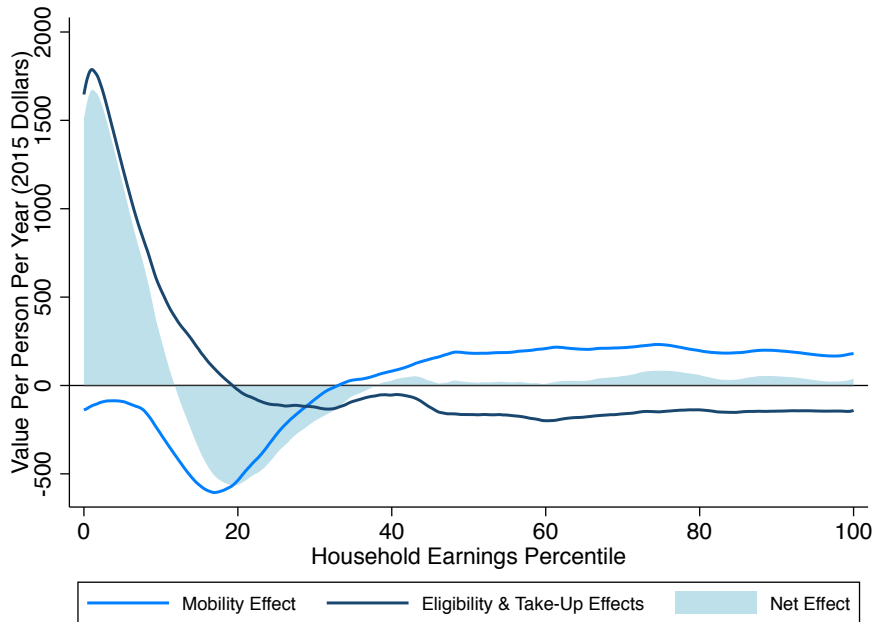
[Back](#)

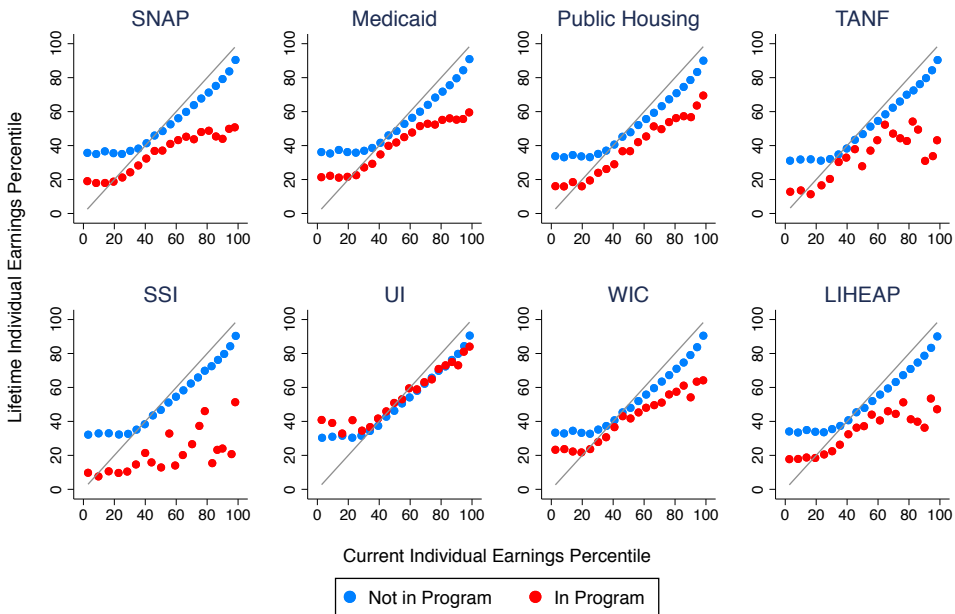


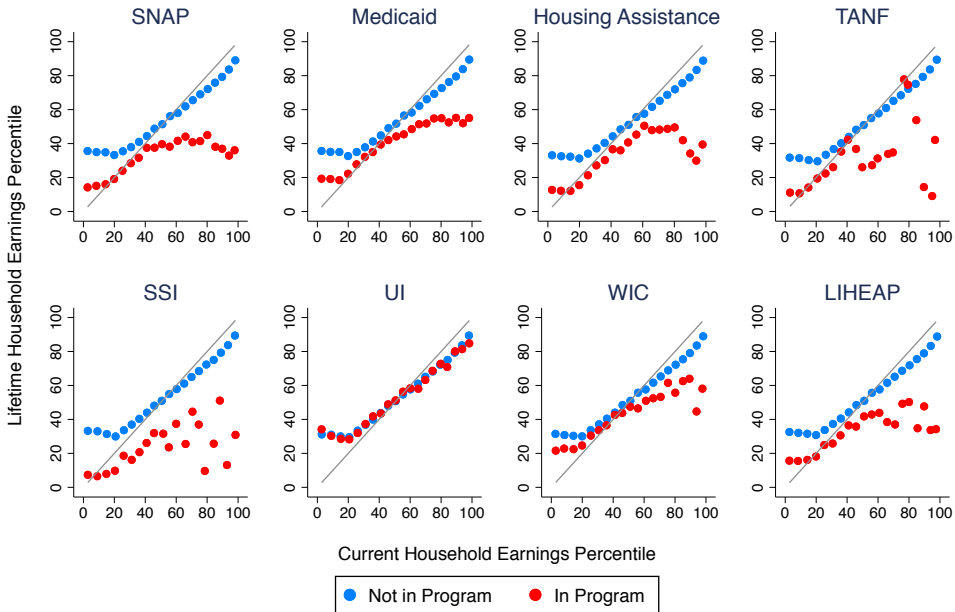












Local Fiscal Externalities from Tax Changes

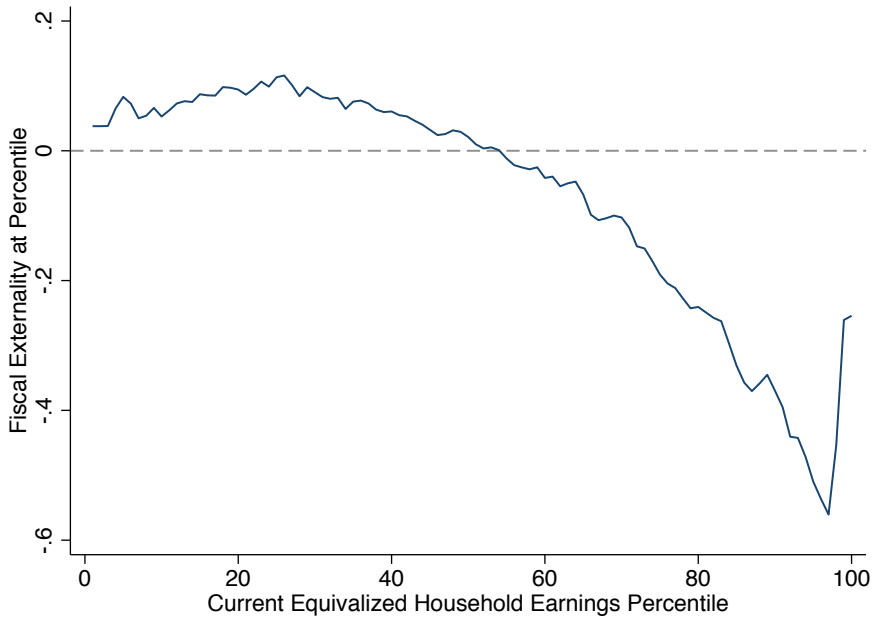
Quantify fiscal externalities from tax change following Hendren (2020)

$$\text{FE}(y) = \underbrace{-\eta_c^P(y) \frac{T(y) - T(0)}{y - T(y)}}_{\text{participation effect}} - \underbrace{\eta_c^c(y) \frac{\tau(y)}{1 - \tau(y)} \alpha(y)}_{\text{substitution effect}}$$

where

- $T(y), \tau(y)$: tax liability and MTR at income level y
- $\eta_c^P(y), \eta_c^c(y)$: compensated participation & intensive labor supply elasticities
- $\alpha(y) = -\left(1 + \frac{yh'(y)}{h(y)}\right)$: local Pareto distributional parameter ($y \sim H(y)$)

→ replicate Hendren's calculations in public-use data for equivalized households



Implementing the Efficiency Metric

- Transfer schedule $t(y, \bar{y})$: Estimate via local linear regression in PSID data

$$\hat{t}(y, \bar{y}) = \hat{\beta}_0(y, \bar{y}) + \hat{\beta}_1(y, \bar{y})y_i + \hat{\beta}_2(y, \bar{y})\bar{y}_i$$

where

$$\hat{\beta}(y, \bar{y}) = \arg \min_{\beta(y, \bar{y})} \sum_i K(y_i - y, \bar{y} - \bar{y}_i) (t_i - \beta_0(y, \bar{y}) - \beta_1(y, \bar{y})y_i - \beta_2(y, \bar{y})\bar{y}_i)^2$$

- Fiscal externality $FE(y)$: Replicate Hendren (2020) in public-use data
 - Pareto parameter $\alpha(y)$: ACS
 - Federal tax schedule $T(y), \tau(y)$: CPS
 - Labor supply elasticities $\eta_c^P(y), \eta_c^C(y)$: Reuse calibrations from paper